

Adapting Communication Styles in Health Chatbot using Large Language Models to Support Family Caregivers from Multicultural Backgrounds

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Abstract

Large language models (LLMs) have enabled the development of increasingly sophisticated AI chatbots, yet these systems rarely consider the cultural context in their interactions. However, individuals from different cultural backgrounds may have distinct communication preferences when interacting with healthcare providers, which can affect therapeutic alliances. This paper presents a technical framework for developing culturally sensitive therapeutic chatbots that systematically adapt to different communication styles. We developed a two-agent system: a therapist agent providing problem-solving therapy, and a communication style modifier agent that adjusts these responses according to six communication styles: Authoritative, Concise, Conversationally Dominant, Informal, Sentimental, and Talkative style. An experimental study demonstrated that our system successfully produced perceivable differences in communication styles, with adapted responses receiving significantly higher style-specific ratings compared to unadapted responses. These results suggest that LLM-based therapeutic chatbots can be effectively modified to exhibit distinct communication styles, laying the groundwork for developing culturally adaptive therapeutic chatbots.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Health care information systems**; • **Computing methodologies** → *Artificial intelligence*.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719711>

Keywords

large language models, health chatbots, cultural adaptation, communication styles, human-computer interaction, user experience, healthcare AI

ACM Reference Format:

Rebekah Lee Baik, Stephanie Lee, Serena Jinchen Xie, Wang Liao, Elina H. Hwang, and Weichao Yuwen. 2025. Adapting Communication Styles in Health Chatbot using Large Language Models to Support Family Caregivers from Multicultural Backgrounds. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719711>

1 Introduction

With the recent advancements in generative large language models (LLMs), the capabilities of conversational artificial intelligence (AI) can be significantly advanced. AI-based chatbots have tremendous potential in healthcare [1]. For example, research has demonstrated that effective use of chatbots can promote mental health and increase adherence to Cognitive Behavioral Therapy [21] [10]. However, it is unclear whether AI-based chatbots, which serve as healthcare assistants or providers, can be equipped with cultural sensitivity and a nuanced understanding of cultural differences [3] to support effective communication that builds therapeutic alliances. In healthcare, building a therapeutic alliance is particularly important for patient compliance with treatment, satisfaction, and health outcomes [6]. Previous studies have shown that LLMs lack an appropriate understanding of cultural differences [9] [2], which may be essential in fostering therapeutic alliances. Furthermore, studies have found that communication styles vary between cultures [13], and such differences in communication style preferences across cultures can impact clinical outcomes. However, despite its significance, there have been limited efforts to incorporate cultural preferences of communication styles into mental health chatbots.

To address this important gap in the literature, this paper aims to (1) identify various communication styles that are the most relevant to developing culturally sensitive LLM-powered chatbots and (2) develop and evaluate therapy bots that incorporate these communication styles.

To address our research goals, we utilized an existing health therapy chatbot, which was developed by the research team to deliver evidence-based psychotherapy for family caregivers. We further developed several therapy bot prototypes with six different communication styles—Authoritative, Concise, Conversationally Dominant, Informal, Sentimental, and Talkative—by using methods such as prompt engineering, parameter-efficient fine-tuning, and transfer learning. We demonstrated that our system successfully produced perceivable differences in communication styles, with adapted responses receiving significantly higher style-specific ratings compared to unadapted responses. We found that adapted chatbot responses received significantly higher style-specific ratings, with the Authoritative and Concise styles showing the greatest improvements. These results highlight the potential of tailoring chatbot communication styles to enhance cross-cultural therapeutic interactions. This paper aims to establish a foundation for enhancing generative AI-based chatbots by integrating communication styles that vary by culture.

1.1 Background and Motivation

While previous studies have shown that a fully-automated mental health chatbot was able to establish therapeutic alliance with human users in a level comparable to human therapists [5], cultural differences can create substantial barriers to developing empathy and therapeutic alliance [18]. Research has shown that patients with Asian backgrounds prefer a more assertive therapist compared to one who does not directly express their own opinions [17]. Additionally, recent studies highlight the importance of cultural factors in shaping the therapeutic alignment of mental health chatbots with users [15].

Mental health interventions that are not culturally adapted can be less effective and may fail to establish strong therapeutic alliances with users from diverse cultural backgrounds. In traditional cultural adaptation methods for interventions, researchers have tailored language to suit different cultures. For example, in a study that adapts a conversational agent for Chinese culture, researchers increased the authority of the conversational agent by using words such as “have to” and “should” [22]. In another study, researchers tailored the messages to be less authoritative and impersonal to align with the cultural values of *personalismo* (i.e., emphasizing warmth and personal connection) [14]. However, the traditional methods of manually tailoring the communication styles for pre-written messages do not effectively adapt to the dynamically generated messages produced by LLM-based tools.

2 Technical Framework

2.1 System Architecture

We used a multi-agent LLM system. Specifically, we built two agents: a therapist agent and a communication style modifier agent. The therapist chatbot was designed to generate responses that emulate a human therapist delivering Problem-Solving Therapy (PST).

The communication style modifier agent then took the responses generated by the therapist agent as input and adjusted the communication style of the responses while preserving the original intent. (see Figure 1)

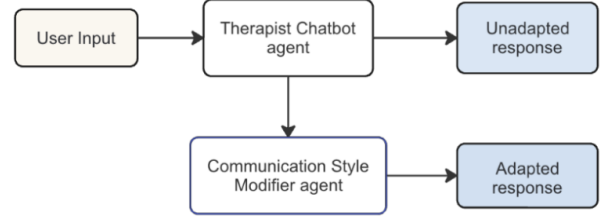


Figure 1: Architecture of Communication Style Adaptation System for Health Chatbot

In previous work, the LLM-based therapist agent was found to be more highly rated overall when the few-shot prompting technique was employed [7]. In this work, we adopted the best-performing prompt structure and utilized examples demonstrating motivational interviewing techniques crafted by a clinical psychologist. (see Table 1)

In designing the communication style modifier prompt, we have used the six identified communication styles that will be further discussed in section 2.2. We experimented with three prompt setups: (1) providing only the communication style name and relying on the LLM to interpret the style, (2) including the communication style name along with a crafted definition, and (3) incorporating the communication style name, the crafted definition, and the corresponding scale items [20]. We used LLM-as-a-judge (Claude-3-opus) [8] to find that the third prompt setup adjusted the style of the responses most consistently and substantially.

We utilized OpenAI’s GPT-4o as our generative model for both the therapist agent and the communication style modifier agent. For the therapist agent, to get responses with higher consistency, we used temperature of 0.7. For the modifier agent, we used the temperature of 1.0.

2.2 Identification of Different Communication Styles

We have conducted a comprehensive literature review on health communication styles across various disciplines, including communication, psychology, anthropology, linguistics, human-computer interaction (HCI), and computer science. From this review, we identified six dimensions of communication styles that are likely to impact the efficacy of AI-based chatbots when interacting with individuals from diverse cultural backgrounds: Authoritative, Concise, Conversationally Dominant, Informal, Sentimental, and Talkative. These communication styles were adapted from the Six-Dimensional Behavioral Model of Communication Styles [20] to better align with chatbot interactions.

Using the scale items from the original model [20], we define these six communication styles as follows: Authoritativeness refers to the tendency to command or demand others in a conversation, without considering their willingness and concerns. Conciseness

Table 1: Prompt Content for Therapist Chatbot and Communication Style Modifier Agents

Agent	Prompt Content
Therapist Chatbot	Your responsibility is to guide the conversation with a caregiver ("user") through the principles of Problem-Solving Therapy (PST) to improve one significant symptom the caregiver is experiencing. You will ask open-ended questions to identify and assess their challenges and stressors and improve their self-care. After you have identified one problem the caregiver can work on to improve their health, generate two concise, achievable, and personalized goals that directly address and support their expressed needs and aspirations. Ensure these goals are not only realistic but designed to inspire and boost the caregiver's motivation. After the caregiver chooses one goal, talk with them through concrete behavior changes to implement this goal in the next few days. Avoid focusing on the care receiver. Remember, your job is to help the caregiver. Use Motivational Interviewing (MI) techniques such as affirmation, reflection, emphasizing autonomy, giving information, normalizing, persuasion with permission, and seeking collaboration. Keeping in mind to cultivate change talk, soften sustain talk, strengthen partnerships, and show empathy. Do not question stack. Below are ideal dialogue examples illustrating how you, the assistant, should evaluate and address the challenges and stressors of caregivers, referred to here as the user, during conversations. {Examples demonstrating MI techniques}
Communication Style Modifier	Please revise the following 'RESPONSE' from a therapist to align better with the {communication style} communication style. This style is characterized by the following definition: {definition} and measured by the survey items: {scale items}. Ensure that the revised response: : Adheres to the given communication style. : Considers the 'CONVERSATION HISTORY' for context. : Asks only one question in the response. CONVERSATION HISTORY: {chat history} RESPONSE to modify: {response}

is characterized by the tendency to use as few words as possible to clearly convey ideas and explain things in a conversation, while avoiding being long-winded. Conversational dominance reflects the tendency to take the lead in a conversation and determine its topics and directions. Informality is the tendency to talk casually and avoid being formal, distant, or stiff in a conversation. Sentimentality represents the tendency to express one's own emotions or display empathic emotional responses to others in a conversation. Lastly, Talkativeness is defined as the tendency to initiate conversations, talk a lot, and avoid silence.

3 Validation and Results

We validated these prototypes through controlled experiments, in which we evaluated human subjects' assessments of the prototypes' communication styles, as well as their preferences for these styles. Additionally, we validated these prototypes using dictionary-based measures of related linguistic markers (Linguistic Inquiry and Word Count, LIWC 2022) (see Appendix B).

3.1 Experimental Setup

We conducted an online experiment using Qualtrics surveys distributed via the Prolific platform. We recruited a total of 100 subjects from the United States and provided them with monetary compensation for their participation. The study employed a mixed design in which communication style was manipulated as a between-subjects factor (with six different styles), while adaptation (adapted vs. un-adapted responses) was manipulated as a within-subjects factor,

allowing each participant to evaluate both adapted and un-adapted versions within their assigned communication style (see Appendix A for details). For experiments, we generated four dialogues for each communication style using four distinct personas, which were verified by a clinical psychologist. Each persona was assigned a fixed response that aligned with its designated characteristics.

Each subject was randomly assigned to evaluate one of six communication styles. Before beginning the evaluation task, subjects were shown a brief tutorial explaining the experimental procedure. Then, the subjects read a dialogue featuring one of four personas. These dialogues contained both un-adapted and adapted responses presented as if they were from therapists, with no disclosure that the responses were generated by an AI chatbot. For both un-adapted and adapted versions of the dialogue, subjects rated the perceived level of the designated communication style using a 5-point Likert scale (1 = Not at all to 5 = Very). For example, for the authoritative style, a rating of 1 means 'not at all authoritative,' while a rating of 5 means 'very authoritative.' After completing the dialogue ratings, subjects viewed the complete dialogue with un-adapted and adapted responses presented side by side. They then indicated their preference between the two "therapist" versions of the AI chatbot. In the final phase, subjects provided demographic information, including their ethnicity and gender.

Throughout the experiment, an attention check was implemented, and data from subjects who failed this check were excluded from the final analysis. After excluding subjects with technical errors, a total of 92 subjects were included in the analysis.

3.2 Communication Style Analysis

The results demonstrate a consistent and significant differentiation between adapted and unadapted responses across all six communication styles (i.e., Authoritative, Concise, Conversationally Dominant, Informal, Sentimental, and Talkative). The results are presented in Figure 2. The adapted responses consistently received higher ratings than their unadapted counterparts, with particularly pronounced differences observed in the Authoritative style (adapted: $M = 4.43$, $SD = 0.86$; unadapted: $M = 2.24$, $SD = 1.13$) and Concise style (adapted: $M = 4.34$, $SD = 0.84$; unadapted: $M = 2.75$, $SD = 1.55$). The adapted versions generally exhibited smaller standard deviations compared to unadapted versions, suggesting greater consistency in participants' recognition of the intended communication styles. Most adapted versions achieved median ratings of 4.0 or 5.0 on the five-point scale, while unadapted versions typically received median ratings between 2.0 and 3.0. This clear separation in ratings, combined with the systematic pattern across all styles and the more concentrated distribution of ratings for adapted versions, provides robust evidence that the AI chatbot successfully incorporated the intended communication styles in a way that was readily perceivable to humans.

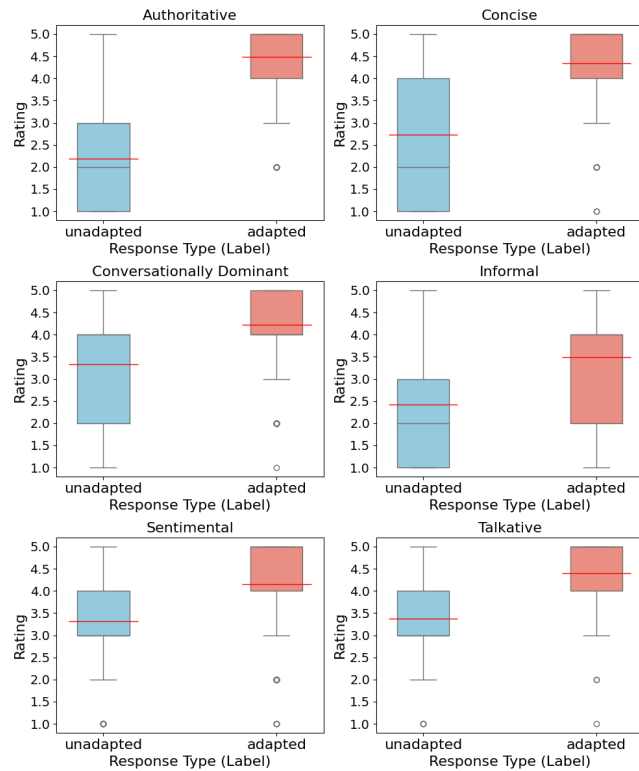


Figure 2: Boxplot of Communication Style Ratings

To statistically validate the differences between adapted and unadapted communication styles, we conducted paired-sample t-tests for each communication style category. This statistical approach was chosen because each participant rated both adapted and unadapted versions, creating naturally paired observations. The

analysis revealed significant differences ($p < 0.05$) between adapted and unadapted responses across all six communication styles. The Authoritative style showed the largest mean difference ($M = 2.19$, $SD = 1.52$), followed by the Concise style ($M = 1.59$, $SD = 1.65$), both yielding particularly strong statistical evidence with p-values well below the conventional significance threshold. The Conversationally Dominant ($M = 0.78$, $SD = 1.51$), Informal ($M = 0.96$, $SD = 1.74$), Sentimental ($M = 0.81$, $SD = 1.47$), and Talkative ($M = 0.96$, $SD = 1.17$) styles also demonstrated statistically significant differences, with all t-statistics indicating strong evidence against the null hypothesis of no difference between adapted and unadapted versions. These results provide robust statistical support for the effectiveness of our adaptation mechanism across all communication styles, with particularly strong effects observed in the Authoritative and Concise categories. The results are summarized in Table 2.

4 Discussion and Future Work

4.1 Key Findings and Implications

Our study yielded several significant findings regarding the effectiveness of communication style adaptation in AI-based therapeutic chatbots. First, we demonstrated that large language models can successfully modify their communication styles in a way that is clearly perceivable to humans, with adapted responses consistently receiving higher style-specific ratings compared to unadapted responses across all six communication styles examined.

The most pronounced differences between adapted and unadapted responses were observed in the Authoritative style (mean difference = 2.19, $p < 0.001$) and Concise style (mean difference = 1.59, $p < 0.001$), suggesting that these communication characteristics are particularly amenable to algorithmic adaptation. The successful implementation of these styles is especially noteworthy given their potential relevance to cross-cultural therapeutic interactions, as previous research has indicated varying preferences for specific communication across different cultural groups.

While all adapted styles showed statistically significant differences over their unadapted counterparts, the magnitude of adaptation success varied. Moderate but significant differences were observed in Informal (mean difference = 0.96), Talkative (mean difference = 0.96), Sentimental (mean difference = 0.81), and Conversationally Dominant (mean difference = 0.78) styles. The consistent achievement of statistical significance ($p < 0.001$) across all styles validates our technical framework's capability to produce reliably distinctive communication patterns.

The smaller standard deviations observed in adapted responses compared to unadapted ones suggest that our adaptation mechanism produces more consistent and predictable communication styles. This consistency is crucial for therapeutic applications, where reliable and stable communication patterns are essential for building therapeutic alliances.

These findings have important implications for the development of culturally sensitive AI-based chatbots for mental health interventions. The demonstrated ability to systematically modify communication styles suggests that therapeutic chatbots could be tailored to match the communication preferences of different cultural groups, potentially enhancing therapeutic alliance and treatment outcomes.

Table 2: Descriptive Statistics and T-Test Results of Communication Style Ratings

Communication Style	Type	Mean	Std	Median	Mean Diff.	Std Dev.	T-Stat	P-Value
Authoritative	Adapted	4.43	0.86	0.86	2.19	1.52	18.73	$4.25 \times 10^{-43}***$
	Un-adapted	2.24	1.13	1.13				
Concise	Adapted	4.34	0.84	0.84	1.59	1.65	12.25	$1.01 \times 10^{-24}***$
	Un-adapted	2.75	1.55	1.55				
Con conversationally Dominant	Adapted	4.21	0.90	0.90	0.78	1.51	5.92	$2.66 \times 10^{-8}***$
	Un-adapted	3.42	1.18	1.18				
Informal	Adapted	3.44	1.27	1.27	0.96	1.74	6.54	$1.08 \times 10^{-9}***$
	Un-adapted	2.47	1.30	1.30				
Sentimental	Adapted	4.13	1.09	1.09	0.81	1.47	6.73	$3.48 \times 10^{-10}***$
	Un-adapted	3.33	0.93	0.93				
Talkative	Adapted	4.35	0.76	0.76	0.96	1.17	10.74	$7.67 \times 10^{-21}***$
	Un-adapted	3.39	0.90	0.90				

Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

This capability is particularly relevant given previous research indicating that communication style preferences vary significantly across cultural backgrounds in healthcare settings.

4.2 Future Directions

Extending our empirical validation that demonstrated significant differentiation between adapted and unadapted communication styles, we will conduct a randomized controlled experiment to test the effects of different communication styles on therapeutic alliance, user engagement, and health outcomes. We plan to recruit 1,400 English-speaking family caregivers from diverse racial and ethnic groups, specifically focusing on the four largest groups in Washington State: White, Black, Hispanic, and Asian Americans [19]. Each participant will be randomly assigned to interact with one of seven prototypes. Six prototypes are adapted to reflect different communication styles, while one unadapted prototype serves as a baseline. We will use block randomization to ensure participants from various cultural backgrounds are evenly distributed among the seven groups. After interacting with a therapy bot, participants will complete surveys to evaluate their perceived therapeutic alliance and other relevant outcomes. By the end of this project, we will have a data-driven understanding of how various communication styles affect the therapeutic alliance, user engagement, and health outcomes and whether cultural backgrounds affect these relationships. Furthermore, we aim to develop a chatbot that ensures cultural sensitivity, enabling it to adapt its communication styles to align with diverse cultural norms and expectations.

4.3 Phase 2 Prototype

In the next phase of the study, we will conduct a randomized controlled experiment to examine the effects of different communication styles on therapeutic alliance and user engagement, as well as the influence of cultural background on these relationships. We have developed six therapy chatbot prototypes incorporating the communication styles adapted and tested in the current study, alongside a baseline chatbot without communication style adaptation. Participants will be assigned to interact with one of the therapy chatbots using block randomization, ensuring that participants from diverse

cultural backgrounds are evenly distributed across the communication style groups. The therapeutic alliance will be directly evaluated using the Short Revised Working Alliance Inventory (WAI-SR) [11]. Additionally, we will measure trust in therapist-patient relationships [4], empathy, and user engagement using a combination of self-reported ratings and automated metrics. This phase of the study aims to provide a data-driven understanding of how communication styles and cultural factors impact the effectiveness of therapy chatbots.

Acknowledgments

This research was, in part, funded by the National Institutes of Health (NIH) Agreement No.1OT2OD032581 and R21NR020634. The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the NIH. This research was also supported by the University of Washington Population Health Initiative Tier 1 Pilot Research Grant Award.

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A Qualtrics Survey

A.1 Tutorial

What You Will Need to Do in This Survey

Each conversation will be formatted as:

[Client Input]

I have tried a mindfulness exercise with breathing.

[Therapist Response]

Great start with mindfulness. How are the exercises helping you?

For each conversation, you will be asked to evaluate the **informal** aspect of the therapist's communication style. You may revisit previous pages to adjust your ratings as needed. Please note there may be an attention check question within the survey, so kindly stay attentive.

Once you have rated all conversations, you will be prompted to fill out a brief demographics section. Did you understand the survey instructions clearly?

Yes

☐

No

☐

Figure 3: Example Tutorial Page for Qualtrics Survey Instructions

A.2 Actual Survey

Conversation # 1-a

[Client Input]

Hi. I'm so tired today, I didn't get enough sleep last night

[Therapist Response]

I'm sorry to hear that you didn't get enough sleep. It sounds really challenging to manage everything when you're feeling tired. How often do you find yourself feeling this way?

Rate the therapist's tendency to talk casually and avoid being formal, distant, or stiff in a conversation (i.e., **informality**) from 1 to 5.

1. Not
informal at
all

☐

2. Slightly
not
informal

☐

3. Neutral

☐

4. Slightly
informal

☐

5. Very
informal

☐

Figure 4: Example Question from the Actual Qualtrics Survey

B Linguistic Inquiry and Word Count (LIWC) Analysis

We utilized the LIWC 22 (Linguistic Inquiry and Word Count) software to analyze the linguistic characteristics of both basic and adapted dialogues across different communication styles. LIWC calculates the percentage of words that fall into various linguistic, psychological, and topical categories, providing quantitative measures of different social, cognitive, and affective processes [12] [16]. The results are presented as standardized LIWC scores (Figure 5, using baseline scores as a reference point for standardization). These standardized scores allow us to identify distinctive linguistic patterns across communication styles. Focusing on categories with $\geq 5\%$ presence, the visualization reveals characteristic linguistic profiles for each communication style. For instance, authoritative communication displayed distinctly lower scores in affect-related categories compared to other styles, consistent with its intended formal and objective nature. The standardized scores demonstrate how each communication style manifests through different linguistic features, providing quantitative validation of our adaptation approach. The patterns observed in the LIWC analysis complement our earlier empirical findings from the user study, offering additional evidence that the AI chatbot successfully incorporated the intended communication styles.

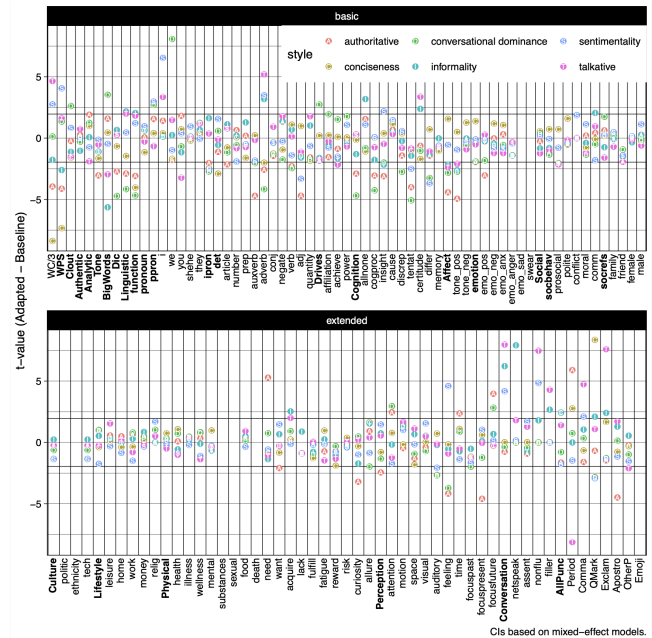


Figure 5: Standardized LIWC scores